

Seat No.	
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Set **P**

**B.Sc. (E.C.S.) (Semester - IV) (New) (NEP CBCS) Examination:
March/April - 2026**

Relational Database Management Systems (G06-DSC1-0402)

Day & Date: Saturday, 18-04-2026
Time: 12:00 PM To 01:30 PM

Max. Marks: 30

Instructions: 1) All questions are compulsory.
2) Figures to the right indicate full marks.

Q.1 Choose the correct alternative.

06

- 1) Atomicity means _____.
 a) All operations are executed or none
 b) Data duplication
 c) Only read operations allowed
 d) Fast execution
- 2) Lost update problem occurs due to _____.
 a) Power failure
 b) Concurrent transactions
 c) Syntax error
 d) Backup
- 3) A shared lock allows _____.
 a) Write Only
 b) delete
 c) Read Only
 d) Both Read and Write
- 4) Heuristic optimization is based on _____.
 a) Cost
 b) Log
 c) Time Stamp
 d) Rules
- 5) GOTO statement transfers control to _____.
 a) Function
 b) Procedure
 c) Label
 d) Cursor
- 6) Trigger is automatically executed when _____.
 a) Procedure runs
 b) Table created
 c) Event occurs
 d) User logs in

Q.2 Answer the following. (Any Three)

06

- a) Define Transaction.
- b) What is concurrency control?
- c) List any 4 advantages of PL/SQL.
- d) What is checkpoint?
- e) What is undo and redo?

- Q.3 Answer the following. (Any Two) 06**
- a) Write a note on ACID property.
 - b) Explain for loop with example.
 - c) Explain package with example.
- Q.4 Answer the following. (Any Two) 06**
- a) Why use lock based protocol? Explain two phase locking protocol.
 - b) What is stored procedure? Explain stored procedure with example.
 - c) Write a note on shadow paging.
- Q.5 Answer the following. (Any One) 06**
- a) What is cursor? Explain different types and attributes of cursor with example.
 - b) What is log? Explain log based recovery techniques in detail.